

Low-rank certification of 2-D PIE stability: the theory behind the demo

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Abstract

Stability of a 2-D PIE is certified by solving a semidefinite program whose matrix variables are far larger than the information they carry: the certificates that exist have rank 2 per block (and exactly $2n$ for n states), while the matrices being searched over have side length in the thousands. This note explains, with a minimum of machinery, why that rank is small, how the demo exploits it by searching only a low-dimensional slice of the positive semidefinite cone, why a certificate found this way can be trusted even though the search is nonconvex, and where the method is known to stop working. Everything quantitative below was measured on the demo's problem family; nothing is asserted from theory alone.

1 The certification problem, and where the cost lives

Given a PIE with operators $(\mathcal{T}, \mathcal{A})$, exponential stability follows from the existence of a PI operator $\mathcal{P} \succeq 0$ (coercive) with

$$\mathcal{T}^* \mathcal{P} \mathcal{A} + \mathcal{A}^* \mathcal{P} \mathcal{T} = -\mathcal{D}, \quad \mathcal{D} \succeq 0. \quad (1)$$

PIETOOLS turns (1) into a semidefinite program in the standard way: $\mathcal{P}$ and $\mathcal{D}$ are parameterized so that positivity is automatic. Concretely, a positive PI operator is written as a quadratic form in a fixed finite basis of simpler operators (monomial multipliers and monomial-kernel integrals): collecting that basis into a column $\mathcal{Z}(s)$,

$$\mathcal{P} = \mathcal{Z}^* Q \mathcal{Z}, \quad Q \succeq 0, \quad (2)$$

so any positive semidefinite *Gram matrix* Q yields a valid $\mathcal{P} \succeq 0$, and (1) becomes a set of *linear* equality constraints on the entries of the two Gram matrices (Q_1 for $\mathcal{P}$, Q_2 for $\mathcal{D}$), obtained by matching the polynomial coefficients of every operator parameter. The decision problem is therefore

$$\text{find } Q_1, Q_2 \succeq 0 \quad \text{s.t.} \quad \mathcal{L}(Q_1, Q_2) = b, \quad (3)$$

with m scalar equality constraints and $\sum_i N_i(N_i + 1)/2$ scalar unknowns, where N_i is the side length of Gram block i .

The side length is $N_i = (\text{states}) \times (\text{basis functions})$, and in two spatial dimensions the basis is what explodes: a basis over the four variables $(x, \theta_x, y, \theta_y)$ at per-variable degree d has $O(d^4)$ elements. At per-variable degree 4 the negativity block of the demo problem already has $N_2 = 744$ for one state, i.e. 277,176 Gram unknowns; four states give 4.43 million.

An interior-point solver does not mind the number of unknowns per se; what it must do at every iteration is form and factor an $m \times m$ system (the Schur complement of the KKT equations). For these LPIs that matrix is *completely dense* — measured on the demo family at $m = 35,800$: every one of the m^2 entries is structurally nonzero — so each iteration costs $O(m^2)$ memory and $O(m^3)$ flops, and m grows like $(\text{states})^2$. Measured consequences: the one-state problem solves in 1707s; the four-state problem needs a 66.8 GiB dense matrix and cannot be attempted on a 64 GiB machine. Memory, not time, is what fails first.

2 The observation: certificates have tiny rank

The matrices being searched over are enormous, but the certificates that exist inside them are not. Measured on the demo family, with every accepted certificate verified as described in Section 5:

- the minimum rank of each Gram block is 2 for one state, and it stays 2 while the basis (and hence N) grows from 424 to 1208;
- across $n = 1, \dots, 6$ replicated states the minimum rank is exactly $2n$ in both blocks — rank grows with *states*, never with *basis size*.

There is a mechanism behind this, and it predicts both halves. A positive operator $\mathcal{P} \succeq 0$ has a square root: $\mathcal{P} = \mathcal{R}^* \mathcal{R}$. If $\mathcal{P}$ is an *integral* operator, its square root can itself be represented in the same operator basis $\mathcal{Z}$, as a small number of rows of coefficients — one row per “square” in the representation. Writing $\mathcal{R}$ with r rows of coefficients $Y \in \mathbb{R}^{N \times r}$ gives $Q = YY^\top$, i.e. $\text{rank } Q \leq r$. The point is that r counts *how many squares the operator needs*, which does not change when the basis is enriched: raising the polynomial degree, or moving from one spatial dimension to two, adds columns to the search space without adding squares to the answer. Adding states does add squares — the square root of an $n \times n$ block operator needs n block rows — which is exactly the measured $2n$.

Two practical corollaries. First, the useful axes are separated: problem difficulty (basis size, spatial dimension, degree) inflates N but not the rank, so the *information content* of a certificate is a handful of numbers even when the SDP has millions of unknowns. Second, the rank tells you what a solver *could* exploit; interior-point methods cannot — they converge to maximum-rank solutions by design (the returned Gram ranks were measured at 417–423 out of 424) — so the structure is invisible unless one goes looking for it.

3 Exploiting it: search a slice, not the cone

Suppose we know (or guess) an $N \times r$ matrix V with orthonormal columns such that a certificate exists of the form

$$Q = VSV^\top, \quad S \in \mathbb{R}^{r \times r}, S \succeq 0. \quad (4)$$

Substituting (4) into (3) leaves a semidefinite program in S alone: $r(r+1)/2$ unknowns instead of $N(N+1)/2$. With $r = 2$ and $N = 744$ that is 3 unknowns instead of 277,140 for that block; the resulting SDP solves in seconds. (In convex-analysis language, (4) restricts the search to a *face* of the positive semidefinite cone; no property of faces beyond the substitution itself is used.)

The step that makes this safe is elementary but load-bearing:

$S \succeq 0$ implies $VSV^\top \succeq 0$, and the constraints $\mathcal{L}(Q) = b$ are linear, so **any** ($S \succeq 0$) satisfying the restricted constraints yields a genuine solution of the original problem (3).

The restriction can only *lose* feasible points; it can never manufacture one.

So a bad guess for V costs a failed attempt, never a false certificate. The converse direction carries no guarantee, and the demo is explicit about this: a rank at which the search fails is reported as “not reached,” never as a proven lower bound.

4 Finding the slice — the weak point, examined

Everything so far is either measured fact or elementary algebra. The genuine weakness of the method is concentrated in one place: *V must be guessed*, and a wrong guess, while never unsafe (Section 3), yields nothing. This section explains how the demo guesses, why that choice is defensible, how it fails, and what the alternatives are.

The proposer: factored descent

The demo finds V by *factored descent* (the Burer–Monteiro method [1]): substitute $Q = YY^\top$ with $Y \in \mathbb{R}^{N \times r}$ directly into the equality constraints and minimize the residual $\|\mathcal{L}(YY^\top) - b\|^2$ over Y by nonlinear least squares (Levenberg–Marquardt), from a handful of random starting points plus a warm start from the previous rank. Two things make this the right tool here:

- the semidefinite constraint disappears — $YY^\top$ is positive semidefinite by construction — leaving an unconstrained problem in $N \cdot r$ variables (a few thousand at $r = 2$, not millions), whose per-iteration linear algebra involves an $(Nr) \times (Nr)$ system;
- nothing in it ever touches an $m \times m$ matrix, so the memory wall of Section 1 never appears, at any problem size.

Y is then orthonormalized to give V , the small restricted SDP (4) is solved exactly, and the verification step decides. The unknown rank is handled by scanning r upward under the same gate, then shrinking the certified face one direction at a time; overshooting r is harmless, since a rank- r face contains every lower-rank certificate built from its directions.

What can be said in its favor — and what cannot

The honest theoretical position is asymmetric. In one direction there is a guarantee of *existence*: any feasible SDP with m equality constraints admits a solution of rank r with $r(r+1)/2 \leq m$ [2], so low-rank solutions ($r \sim \sqrt{2m}$, here 100–270) always exist. In the other direction there is a guarantee about the *search*: when r is taken large enough that $r(r+1)/2 > m$, the factored problem has, for generic data, no spurious local minima — descent finds a global solution [3]. The demo operates far below that threshold ($r = 2$ against m in the thousands), where no such guarantee applies and stalls in local minima are possible in principle. What can be claimed instead is empirical, and it was measured rather than assumed:

- at the ranks where certificates exist, the descent found them at the *first* random seed in most measured cases, and within three seeds plus a warm start in the rest;
- its characteristic failure is a *near-miss*, not divergence: a residual settling just above the acceptance gate. One measured case stalled at 1.6×10^{-6} at rank 2 on a problem where a rank-2 certificate provably exists (it was later exhibited by a structural construction); the first off-family test showed the same signature.

That failure mode is why the architecture is a *proposer–checker* split, and the split does real work: the nonconvex step only has to find the right r -dimensional *subspace*, not the right point in it — the restricted SDP then repairs the point exactly within that subspace, and the verification step (Section 5) is the sole arbiter. A stalled descent therefore costs minutes and produces the report “rank r not reached,” which is the correct statement: with this proposer, a failed rank is never evidence of a rank floor.

For a new operator the discovery step is a one-time cost (measured 11–25 minutes on two cold problems) — cheaper than a single full interior-point solve, but not free. Everything after it is.

Alternatives, and their measured status

Guessing V is a heuristic, so the natural question is what else could supply it. Each of the following was either measured on this problem class or is flagged as untested:

Truncating the interior-point solution. The obvious first idea, and it fails for a structural reason: interior-point methods converge to the *maximum*-rank point of the solution set (measured Gram ranks 417–423 of 424), and the leading eigenvectors of such a point are not the low-rank subspace. Faces truncated from full solves certified nothing on the demo family — including in a control experiment where the restriction was made generous enough that success was expected.

Trace minimization. The standard convex surrogate for low rank (minimize $\text{tr } Q$, then truncate). It supplied usable faces on related 1-D problems, but on the demo family it was measured to fail: the solver exits with a numerical error and the trace objective crushes the small Lyapunov block to noise, so its truncated faces fail at every rank tried. Worth attempting on a new problem — it is deterministic — but not to be relied on.

Structure: replication — not transfer. When the problem has replicated states, Section 6 removes the guess entirely: the $n = 1$ face is constructed, not searched, at every larger n . That is where structural reuse ends. Reusing a face across a *physics* change is measured to fail and has been abandoned as a goal: on the demo family, the $\lambda = 2$ face gives an $O(1)$ operator residual already at $\lambda = 1.9$, while a cold search re-certifies the new problem at the *same* rank — the certificate’s rank survives the parameter change, its subspace does not. Every new operating point pays its own discovery.

Physics-informed construction. The mechanism of Section 2 says the face *is* the operator square root written in the basis. When a candidate Lyapunov operator is known in closed form, V can be written down rather than searched: on a 1-D beam example, injecting the known energy $\text{diag}(1, c)$ reproduced the certificate to machine precision. For PDEs with a known energy functional this is the cheapest route of all, and it is the most promising direction for removing the guess in general.

Purpose-built low-rank SDP solvers. Solvers that natively exploit low rank exist — manifold-optimization methods and sketching-based conditional-gradient methods among them. On 1-D analogues of these LPIs they were measured and did not overtake the pipeline above before the problem sizes of interest; they have not been re-evaluated in 2-D, where the balance could differ. They are alternatives to the *whole* pipeline rather than to the guess of V .

5 Trust the check, not the search

Because the proposer is nonconvex and the solver is numerical, no certificate is accepted on the word of either. Acceptance is a direct check of the original operator identity: reconstruct the PI operators $\mathcal{P}$ and $\mathcal{D}$ from the candidate Gram matrices and evaluate

$$\frac{\max |\mathcal{T}^* \mathcal{P} \mathcal{A} + \mathcal{A}^* \mathcal{P} \mathcal{T} + \mathcal{D}|}{\max |\mathcal{T}^* \mathcal{P} \mathcal{A} + \mathcal{A}^* \mathcal{P} \mathcal{T}|} < 10^{-6}$$

where the maxima run over the coefficients of *every* parameter of the resulting operators (all 36 parameter cells of a 2-D PI operator — checking a subset was measured to pass garbage), together with positive semidefiniteness of every Gram block. Both halves are necessary: a point satisfying the equalities to 10^{-11} but indefinite was measured, as was the reverse. In this sense every certificate the demo reports is approximate — exact up to a 10^{-6} relative residual — which is the same standard the underlying SDP solvers work to.

6 The structure bonus: states replicate for free

The Gram basis produced by PIETOOLS is organized in groups (one per kind of basis operator), with the state index nested *inside* each group. For a family of n replicated states this has a striking consequence: if X_1 is a certified one-state Gram, then placing $I_n \otimes (\cdot)$ *within each group* produces an n -state certificate with no optimization at all — write it down, verify it, done. (The naive global $I_n \otimes X_1$, ignoring the group structure, is wrong and fails; the demo’s `pielr_tensor` performs the correct per-group construction and checks the group layout against the actual Gram sizes before using it.)

Measured: the $n = 1$ certificate replicates through $n = 6$ with the residual unchanged to four digits (9.291×10^{-7}), including sizes where the direct solve is memory-infeasible. This is why the demo’s “solve” column is seconds at every n : after one discovery at $n = 1$, larger n requires only assembly and verification.

7 What it buys, and where it stops

n	Gram unknowns	setup (s)	certify (s)	residual	direct solve
1	277,176	5.5	1.6	9.3×10^{-7}	1707 s
2	1,107,952	26.4	6.2	9.3×10^{-7}	~8.6 h (est.)
3	2,492,328	68.5	16.0	9.3×10^{-7}	~98 h (est.)
4	4,430,304	278	47	9.3×10^{-7}	infeasible (66.8 GiB)

Setup (assembling the LPI) is now the dominant cost: the cone side has been removed from the scaling path, and what remains is program construction.

Known limits, all measured rather than conjectured:

1. **Multipliers cost their sum-of-squares length.** The mechanism of Section 2 is about integral operators. If the certificate is *forced* to reproduce a multiplication operator $a(s)$ — as in weighted inequalities, or Lyapunov weights for variable-coefficient PDEs — then the corresponding Gram sub-block must represent a as a sum of squares, and its rank is exactly the number of squares a needs. In one spatial variable that number is at most 2 (every nonnegative univariate polynomial is a sum of two squares), so the effect is benign. In two variables it is unbounded in degree (measured: $a = 1 + x^2 + y^2$ forces rank 3), and a weight that is positive on the domain but *not* a sum of squares makes the problem infeasible outright at these settings. Pure PDE stability problems dodge all of this structurally: their negativity operator has no multiplication part whenever $\mathcal{T}$ and $\mathcal{A}$ have none, which holds for every 2-D example shipped with PIETOOLS.
2. **Low rank is a property of certificates with margin.** As the operating point approaches the boundary of what the LPI can certify at all, the minimum rank climbs toward full

(measured: floor 3 at half the certifiable boundary, 6 at 0.8, none found ≤ 6 at 0.95). Borderline certificates are not compressible.

3. **The measured laws are regularities, not theorems.** Rank 2, the $2n$ law, and the replication property are measurements on the reaction–diffusion family plus two PIETOOLS library examples, at the direct-form stability LPI with no Positivstellensatz terms. The API exists precisely so that other systems can be probed; the first off-family datapoint (an anisotropic variant) certified at rank [3 3] with rank 2 not reached — consistent with the picture, but a reminder that the constants are problem-dependent.

References

- [1] S. Burer and R.D.C. Monteiro, “A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization,” *Mathematical Programming* 95(2):329–357, 2003.
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- [3] N. Boumal, V. Voroninski, and A.S. Bandeira, “The non-convex Burer–Monteiro approach works on smooth semidefinite programs,” *NeurIPS* 2016; deterministic guarantees in *Comm. Pure Appl. Math.* 73(3):581–608, 2020.